ECN 711
Problem Set #2

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Due date: Sunday, September 14th, 2025.

1. Consider an economy with two goods and $n \geq 2$ people. Endowments of both goods are given by $\omega^h = (\bar{\omega}_1^h, \bar{\omega}_2^h)$, $h = 1, ...n$.

Individual preferences are the same for all individuals and represented by the utility function

$$u(x^h) = \ln(x_1^h) + \ln(x_2^h),$$

$h = 1, ...n$.

1. Find the equilibrium price $p = p_1/p_2$. Show that it depends only on the *aggregate* endowments of both goods.
2. Find the equilibrium allocations.

2. Consider an economy with two goods and $n = 2$ people. Endowments of both goods are given by $\omega^h = (\bar{\omega}, \bar{\omega})$, $h = 1, ...n$.

Individual preferences are the same for all individuals and represented by the utility function

$$u(x^h) = \sqrt{x_1^h} + \sqrt{x_2^h},$$

$h = 1, ...n$.

1. Find the equilibrium price $p = p_1/p_2$. Find the equilibrium allocations.

2. Suppose now that endowments are given by $\omega^h = (\bar{\omega}_1, \bar{\omega}_2)$, $h = 1, ...n$. Find the equilibrium price.

3. Consider an economy with two goods and $n \geq 2$ people. Endowments are the same for all individuals: $\omega^h = (\bar{\omega}, \bar{\omega})$, $h = 1, ...n$.

Individual preferences are represented by the utility function

$$u^h(x^h) = \ln(x_1^h) + h \ln(x_2^h),$$

$h = 1, ...n$.

1. Find the equilibrium price $p = p_1/p_2$. What is the effect of the equilibrium price p of an increase in the number of people, n? Explain.
2. Find the equilibrium allocations.

4. Consider an economy with two goods and $n = 2$ people. Endowments of both goods are: $\omega^h = (\bar{\omega}_1, \bar{\omega}_2)$, $h = 1, 2$.

Individual preferences are the same for all individuals and represented by the utility function

$$u(x^h) = (x_1^h)^a (x_2^h)^{1-a},$$

$h = 1, 2$, and $a \in (0, 1)$.

There is a government in this economy that taxes the consumption of good 2 at the rate $\tau \in (0, 1)$ to finance its consumption. Government consumption does not provide any utility to individuals.

1. Define a competitive equilibrium.
2. Find the equilibrium price $p = p_1/p_2$. Does taxation change the price relative to the case of no taxation at all? Explain.

3. Find equilibrium allocations.

5. (Bonus) Consider now a variation of problem 1 in which the utility function is given by

$$u(x^h) = \ln(x_1^h) + \ln(x_2^h - k),$$

$h = 1, ...n$. k is a 'small' and positive parameter, satisfying $k < \bar{\omega}_2^h$, for all h.

Solve for the equilibrium price, $p = p_1/p_2$. What are the effects on the equilibrium price of a lower value of k?

Suppose now that the endowments of both goods increase in the *same* proportion for *all* individuals. What are the effects on the equilibrium price? Explain.